

Offline Flyback SMPS — Complete Design Session Notes

1. Project Intent and Scope

This document records the complete design, bring-up, validation, and refinement of an offline isolated flyback SMPS. The objective was to design a fully functional power supply from first principles, including custom magnetics, closed-loop regulation, protection, and measurement, suitable for professional portfolio and job presentation.

2. Electrical Specifications

Topology: Isolated flyback converter

Input: 120 VAC only (rectified to ~170 VDC bulk)

Prototype Output: 5.00 V @ ~0.3 A

Planned PCB Output: 5 V @ 1 A

Controller: TinySwitch (TNY285 class)

Feedback: TL431 + optocoupler

Isolation: Transformer + optocoupler

Auxiliary Winding: None

3. Primary-Side Architecture

The TinySwitch controller was used without an auxiliary winding, relying on internal drain-derived biasing. The EN/UV pin was populated with a high-value resistor from the rectified HV bus to enable startup and UVLO behavior. Regulation is achieved by optocoupler transistor modulation of the EN/UV pin.

4. Custom High-Frequency Transformer

A fully custom, hand-wound high-frequency transformer was designed without reference designs. Correct turns ratio, polarity, magnetizing inductance, and leakage behavior were validated indirectly through immediate stable regulation, controlled drain stress, and acceptable ringing behavior.

5. Clamp and Early Failure

Initial operation without a leakage clamp resulted in controller failure due to uncontrolled drain overvoltage. This failure confirmed the necessity of a clamp network rather than indicating a design flaw. After implementation of an RCD clamp, no further failures occurred.

6. RCD Clamp Validation

The RCD clamp successfully limited drain overvoltage. Drain-to-source voltage was measured using a 500× differential probe at 120 VAC input. Maximum observed peak drain voltage was approximately 330–345 V, providing substantial margin to the MOSFET rating.

7. Drain Waveform Characterization

Multiple captures showed consistent V_{p-p} values (~ 300 V actual) across load conditions. Absolute peak voltage was determined separately by zooming on the highest excursion, not by V_{p-p} alone. This confirmed correct understanding of waveform interpretation and device stress assessment.

8. Leakage Ringing Analysis

Leakage ringing was observed at approximately 340–390 kHz. The ringing decayed within a few cycles, indicating acceptable leakage inductance and effective damping. The observed frequency is consistent with leakage inductance resonating with drain capacitance.

9. Secondary-Side Power Path

A Schottky rectifier was used for the 5 V output. Initial output capacitance was minimal, resulting in measurable ripple and HF spikes. This behavior was expected and identified as an output filtering issue rather than instability.

10. Feedback Loop Design

The TL431 provided error amplification with a resistor divider setting 5.00 V output. Divider dissipation was calculated in the milliwatt range, making 1/8 W or 1/4 W resistors sufficient. A small compensation capacitor (~ 1 nF) ensured loop stability.

11. Load Regulation Testing

The supply was tested at 50 mA, 100 mA, 227 mA, and near 300 mA. Output voltage remained tightly regulated at 5.00 V with no audible noise or instability. Startup and load transitions were benign.

12. Output Noise Analysis

Two noise components were identified: high-frequency spikes from diode recovery and leakage inductance, and low-frequency ripple from energy packet delivery. Planned mitigation includes a secondary RC snubber, increased bulk capacitance, and addition of μ F-range X7R ceramic capacitors.

13. Efficiency Measurement Strategy

True efficiency is defined from AC wall input to DC output. Preferred measurement uses real AC input power via wattmeter or scope-based $v(t)i(t)$ integration. DC bus power measurement is used as a secondary metric to characterize converter-stage efficiency.

14. USB Charger Adaptation

To function as a USB charger, the design requires appropriate output connectors and port identification. USB-C operation requires 5.1 k Ω pull-down resistors on CC1 and CC2 to advertise default power.

15. Middlebrook Loop Injection Plan

Loop gain measurement is planned via Middlebrook injection in the optocoupler-to-EN/UV path. Sweeps from ~ 10 Hz to tens of kHz will be used to extract crossover frequency and phase margin.

16. Readiness for PCB Implementation

The design is electrically complete pending secondary snubber installation, output filtering refinement, and efficiency characterization. PCB planning includes optional clamp/snubber footprints, tight current loops, creepage control, and test points.

17. Engineering Assessment

This project demonstrates end-to-end offline SMPS design capability including magnetics, control loops, protection, waveform interpretation, and professional measurement discipline.

18. Key Takeaway

The supply operated correctly on first complete assembly due to correct topology, magnetics, and feedback design. This establishes a repeatable SMPS design workflow suitable for rapid future designs.